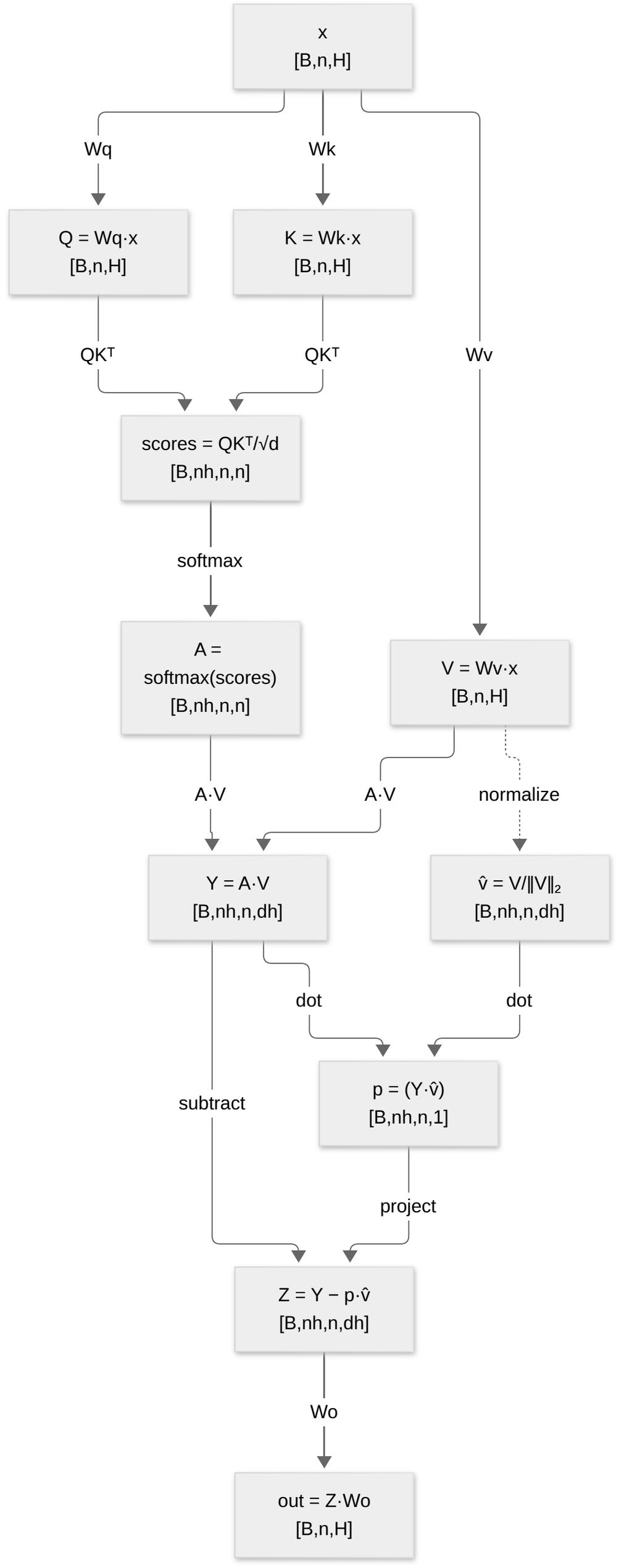

x
[B,n,H]
Wq
Wk
Wv
Q = Wq·x
[B,n,H]
K = Wk·x
[B,n,H]
QKᵀ
QKᵀ
scores = QKᵀ/√d
[B,nh,n,n]
softmax
A =
softmax(scores)
[B,nh,n,n]
V = Wv·x
[B,n,H]
A·V
A·V
normalize
Y = A·V
[B,nh,n,dh]
v̂ = V/‖V‖₂
[B,nh,n,dh]
dot
dot
p = (Y·v̂)
[B,nh,n,1]
subtract
project
Z = Y − p·v̂
[B,nh,n,dh]
Wo
out = Z·Wo
[B,n,H]